



KA-7421

**Integrated M.C.A. (Sem-I)
Examination, December-2024**

IMCA - 101T

Programming Fundamentals Using C

Time allowed : Three Hours

Maximum Marks : 70

Minimum Marks : 28

Note : This question paper contains two sections as under:

SECTION - A

- 1**
- (i) How do you take user input in C?
 - (ii) What is the difference between = and == in C?
 - (iii) What is a pseudocode?
 - (iv) What is a keyword? Give two examples.
 - (v) What is the syntax of a For loop?
 - (vi) What is an infinite loop?
 - (vii) What is the difference between one dimensional and a two - dimensional array?
 - (viii) What is a NULL pointer?
 - (ix) What is the purpose of # include in C?
 - (x) How do you declare a 'structure' in C?

SECTION - B

UNIT-I

- 2**
- (a) Explain the structure of a 'C' program with an example.
 - (b) What is an algorithm? Explain its characteristics and importance in 'C' programming.

OR

- (a) Draw a flowchart to check if given number is EVEN or ODD.
- (b) What are the various data types available in 'C' language? Give their importance.

UNIT-II

- 3
- (a) Write a program to implement a calculator in C language, using SWITCH-CASE statement and different operators.
 - (b) What do you understand by operator precedence in 'C' language? Explain its utility with suitable example.

OR

- (a) Explain the different operators available in 'C' programming. Also, give their significance.
- (b) Write a 'C' program to find the maximum of two numbers using the 'ternary(?:)' operator.

UNIT-III

- 4
- (a) Write a 'C' program to print the fibonacci series using a loop.

- (b) Give comparisons between FOR loop, WHILE loop and DO-WHILE loop with suitable examples.

OR

- (a) Explain the purpose of break and continue statements in 'C' programming, with examples.
(b) Write a 'C' program to print the following:

```
      *
    *   *
  *   *   *
*   *   *   *
  *   *   *
    *   *
      *
```

UNIT-IV

- 5 (a) Explain the differences between global and local variables used in functions in C programming. Give suitable examples.
(b) What do you understand by pointers? Give their significance in memory management.

OR

- (a) Write a C program to illustrate sorting using arrays.
- (b) What do you understand by arrays? Explain the different types of arrays with examples.

UNIT-V

- 6** Write short notes on:
- (a) Nested structure
 - (b) Array of structure
 - (c) Enum

OR

Write short notes on:

- (a) Arrays within structures
 - (b) Union
 - (c) Differentiate structure and union
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